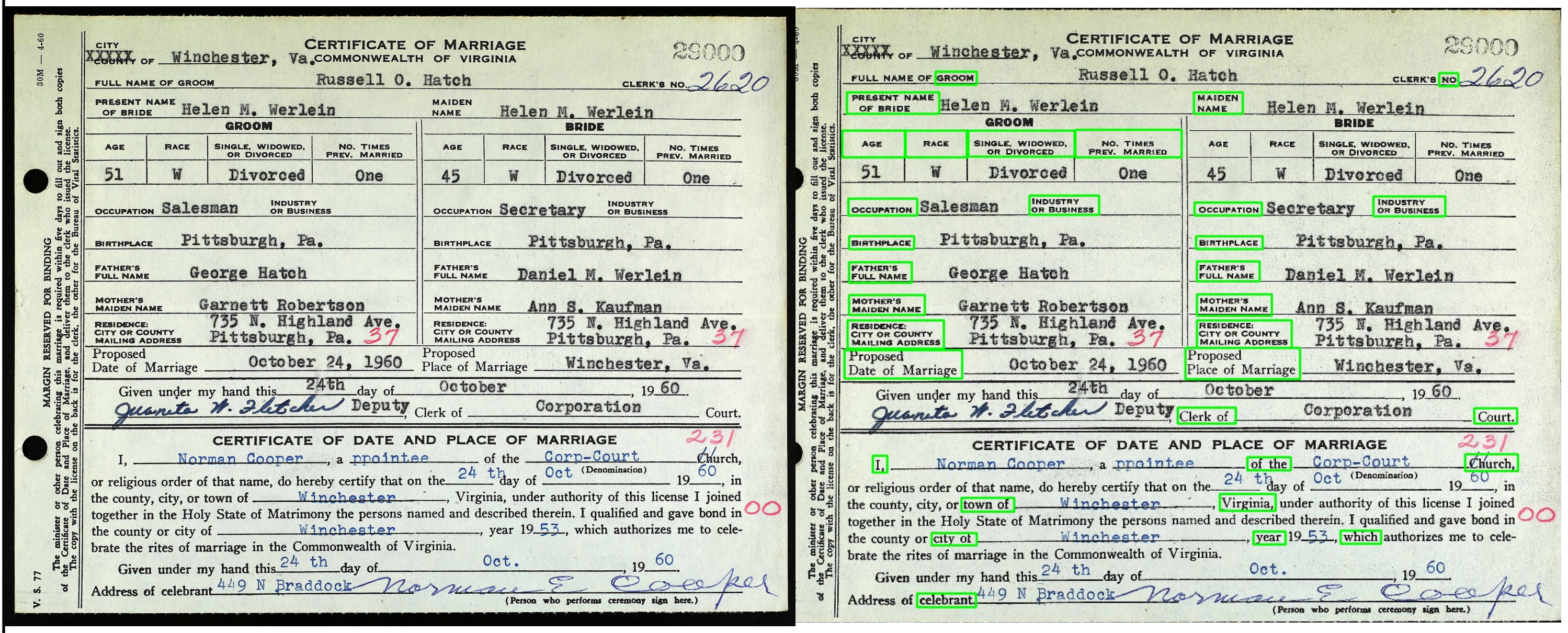


BACKGROUND

Objective: Automatic extraction of historical genealogical data

Challenge: Historical documents have many formats and individual handwriting can vary

Focus: Historical records, such as birth, marriage, and death certificates



GOAL

Use **Object detection** and **Optical Character Recognition** to automate the process

Simple: Limit labeled data required

Replicable: Design should be easily replicable and transferable to similar problems

Accuracy: Aiming for over 90% accuracy for dependability

DATASET

Sourced: Ancestry.com to label

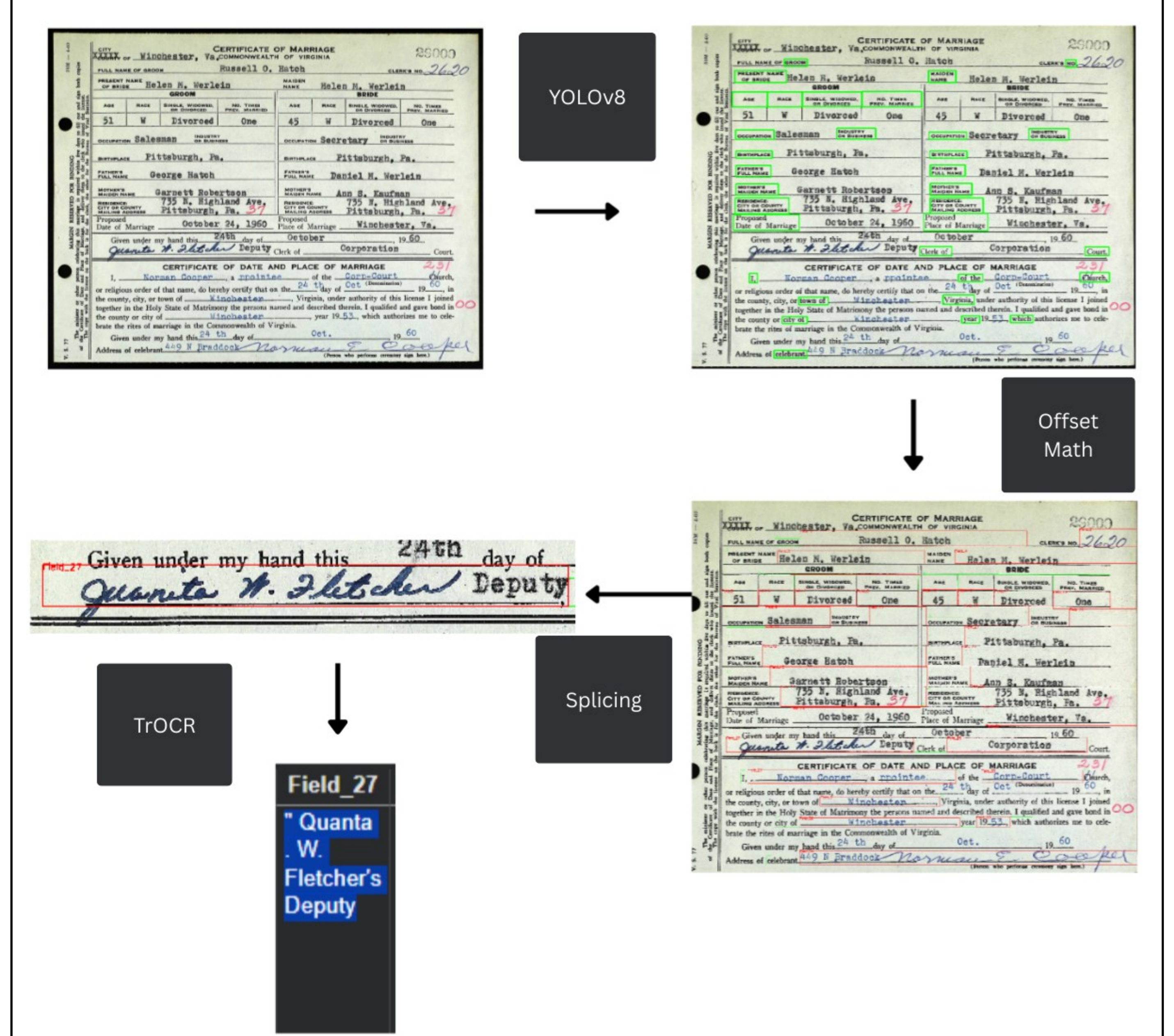
Dataset Name: Virginia. Marriage Certificates

October 1960

Size of dataset: 20 Images (15:5)

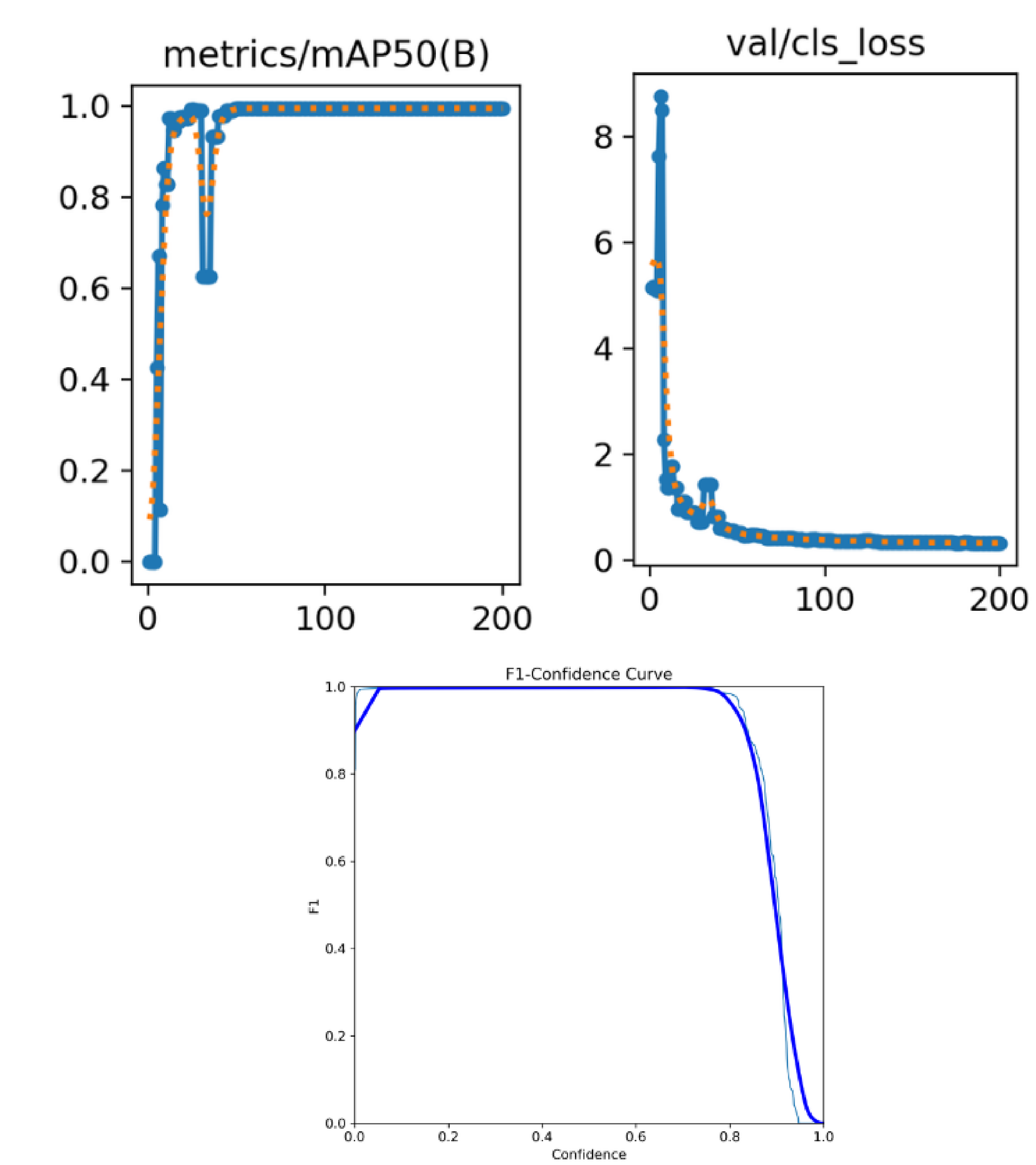
Augmentation: YOLOv8 defaults except for lr-flip and scale which were turned to 0.0

METHODOLOGY



RESULTS

After 200 epochs we have the following results:



FUTURE PLAN

- Greatly improve accuracy of OCR models
- Improved ROI extraction with better equations to gather offset
- Test the same system on brand new dataset to confirm findings

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